U.S. Traffic Accident Modeling: Final Report

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Machine Learning 2

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Introduction and Literature Review

Traffic accidents represent a major public safety concern in the United States, causing significant costs, injuries, and fatalities each year. Understanding the factors that contribute to accident severity is important for improving road safety and informing transportation policy. Advances in machine learning have enabled researchers to analyze large traffic datasets and identify patterns that traditional statistical models may overlook. By leveraging environmental, temporal, and roadway variables, machine learning predictive models can uncover the conditions that increase the likelihood of serious crashes. These insights could help policymakers and engineers design more effective safety interventions.

Recent research has applied a variety of modeling approaches to forecast accident trends and analyze accident severity. For example, Bakis et al. (2025) examined traffic accident data from Batman, Turkey between 2013 and 2022 to predict future accident trends. The study applied multiple predictive models including artificial neural networks, ARIMA, and CNN models. The results showed that machine learning models were capable of producing reasonably accurate predictions of accident trends, though performance varied depending on the model and parameters used. The authors also found that accident rates and related injuries are expected to increase in the future if no preventative measures are taken. This highlights the importance of predictive modeling for traffic safety planning.

Similarly, Al-Masaeid and Khaled (2023) investigated traffic accident prediction models using accident data from Jordan covering the period between 1995 and 2020. Their study compared several modeling approaches, including regression models, artificial neural networks, and ARIMA time-series models to predict the number of accidents, injuries, and fatalities. Model performance was evaluated using metrics such as RMSE, MAE, and R^2. The results indicated that artificial neural networks produced the most accurate predictions among the tested models, followed by ARIMA models and traditional regression techniques. The findings suggest that neural networks are particularly effective at modeling complex nonlinear relationships present in traffic accident data.

Together, these studies demonstrate the growing role of data-driven approaches in traffic safety research. However, several gaps remain in the literature. First, many studies draw on accident data from specific regions or countries which may limit the generalizability of their findings to different driving environments and/or infrastructure contexts. Second, studies that do

apply time-series modeling tend to focus narrowly on forecasting total accident counts, without examining the underlying environmental and situational factors that increase accident severity. Third, probabilistic approaches such as Bayesian classification seem underutilized in this space. They can be used to quantify uncertainty in predictions and identify the relative importance of contributing factors.

The present study addresses these gaps using a large-scale US traffic accident dataset from 2016-2023 with approximately 7.7 million records. We applied a two-model approach: the first being an OLS time-series model that decomposes daily accident frequency into trend, seasonal, and weekly components (including the impact of COVID-19 on accident rates). The second model is a Bayesian categorical regression model built in R using *brms* that predicts accident severity (1-4) from environmental and situational features such as temperature, visibility, humidity, time of day, and weather conditions. Together these models aim to both capture high level trends in accident frequency and provide interpretable predictions of accident severity at the individual level.

ML Methodology and Data Collection

The dataset used in this study is the US Accidents dataset, sourced from Kaggle and originally compiled from multiple traffic incident reporting APIs covering 49 US states. The dataset contains approximately 7.7 million accident records spanning February 2016 through March 2023, making it one of the largest publicly available traffic accident datasets. Each record includes a set of features covering accident timing, geographic location, weather conditions at the time of the incident, road infrastructure characteristics, and a severity rating on a scale of 1 to 4, where 1 represents minimal impact and 4 represents the most severe. The breadth of this dataset makes it well suited to both temporal trend modeling and severity classification, as it provides both longitudinal coverage across seven years and a wide range of environmental and situational predictors.

Because the full 7.7 million record dataset is computationally intensive, a random sample of 10,000 records was used for both models and throughout this project. The core preprocessing/cleaning strategy was developed in Python and later translated to R for the Bayesian model. From the full set of 46 columns, nine features were selected as most relevant to the modeling objectives: Start_Time, Severity, Temperature(F), Humidity(%), Visibility(mi), Weather_Condition, Sunrise_Sunset, State, and Distance(mi). Rows with missing values in

Start_Time or Severity were dropped entirely, as these are critical features for the models. Remaining missing values in numeric columns were imputed using column medians, and missing categorical values were filled using the column mode. Timestamps were parsed and decomposed into derived temporal features including year, month, hour, and day of week, which were used as inputs to both models.

The first model treats accident frequency as a time series problem, modeling daily accident counts in Python using three approaches: an OLS decomposition, an AR(7), and a SARMA(1,1)(1,1)$_7$. Before model fitting, the raw data was aggregated into daily accident counts and timestamps were parsed to extract temporal features. Missing numeric values were imputed with column medians. For the OLS model, the time axis was normalized to a zero-to-one scale for numerical stability and four additive components were constructed as features: a quadratic term for long-term growth, three Fourier harmonic pairs for annual seasonality, three Fourier harmonic pairs for weekly seasonality, and a binary indicator for the COVID-19 lockdown window in 2020. The quadratic trend was chosen over a linear one to accommodate the accelerating growth in accident reporting observed across the dataset span.

The AR(7) model takes a different approach, predicting each day's count directly from the previous seven days of observed counts. Using one full weekly cycle as the lag window captures the strong day-of-week pattern in the data without requiring explicit seasonal features. The SARMA(1,1)(1,1)$_7$ extends this further by incorporating both autoregressive and moving average terms at lag 1 and lag 7, allowing the model to account for both recent momentum and how unexpected deviations from the prior week tend to correct over time. Because MA terms cannot be fit by OLS directly, the Hannan-Rissanen two-step method was used: a long AR(20) is first fit to approximate the unobserved innovations, and those residuals are then used as proxies for the MA terms in a second OLS pass. All three models were fit using the OLS normal equations and evaluated on a chronological 80/20 train/test split using R^2, MAE, and RMSE.

The second model was a Bayesian built in R using the *brms* package. The goal is to predict accident severity (1–4) from environmental and situational features. Before model fitting several preprocessing steps were applied. The raw hour of accident was discretised into five time-of-day periods: Night, Morning, Afternoon, Evening, and Late Night. Raw weather condition strings were collapsed into five interpretable categories (Clear, Cloudy, Rain, Fog, and Snow) using keyword matching, with Rain expanded to include thunderstorm and T-storm

conditions. Observations with unclassifiable weather conditions were excluded from analysis. A binary indicator for nighttime conditions was derived from the Sunrise_Sunset variable. Continuous predictors like temperature, humidity, visibility, and accident distance were standardized to zero mean and unit variance prior to modeling, ensuring coefficient estimates are comparable across predictors and that the specified priors are meaningful on a common scale.

The dataset had substantial class imbalance for the Severity feature. Severity 2 and 3 accounted for the majority of observations. To mitigate the tendency of the model to predict only the dominant class, a downsampling strategy was applied. Severity 2 and 3 were each capped at 800 observations, while Severity 1 and 4 were kept in full. This yielded a working dataset of 1,888 observations. An 80/20 train/test split was then applied.

The model was trained to predict accident severity using seven predictors: temperature, humidity, visibility, accident distance, time of day, weather conditions, and whether the accident occurred at night. Prior beliefs about the model parameters were specified before seeing the data, reflecting an assumption that most predictors would have modest effects. Large or extreme effects were permitted but not expected. The model was fit using four Markov Chain Monte Carlo (MCMC) chains which works by repeatedly drawing plausible parameter values from the posterior distribution until those draws stabilize into a reliable estimate. We used the categorical likelihood with a logit link. Performance was evaluated on a 20% test set using posterior predictive inference, with predictor influence assessed via conditional effects plots.

ML Results and Interpretation

Time Series

All three time series models were evaluated on the final 20% of the daily accident count series, approximately 448 days. None performed exceptionally well in absolute terms, which is expected given that the 10,000-row sample produces daily counts averaging around four accidents per day. This creates a high noise-to-signal ratio that makes temporal patterns difficult to recover reliably. That said, there are meaningful differences across the three approaches, and performance improves consistently as the models become more sophisticated.

Model	Test R^2	MAE	RMSE
OLS Decomposition	−0.046	2.258	2.795
AR(7)	0.104	1.986	2.587

SARMA(1,1)(1,1)$_7$	0.155	1.944	2.512

Table 1

The OLS decomposition struggled the most. Its test R^2 was negative, meaning it performed worse than simply predicting the daily mean. The issue is the quadratic trend component, which fits the training window reasonably well but extrapolates too aggressively into the test period where accident counts are both higher and more volatile. Despite this, the model recovers interpretable structure from the data. Figure 1 shows a steady upward trend rising from approximately 2.2 to 6.5 accidents per day over the full 2016 to 2023 span, a consistent annual seasonal cycle, and a visible COVID-19 disruption in early 2020. The estimated drop during the lockdown window was approximately 0.7 fewer accidents per day, which aligns with the sharp reduction in travel during that period.

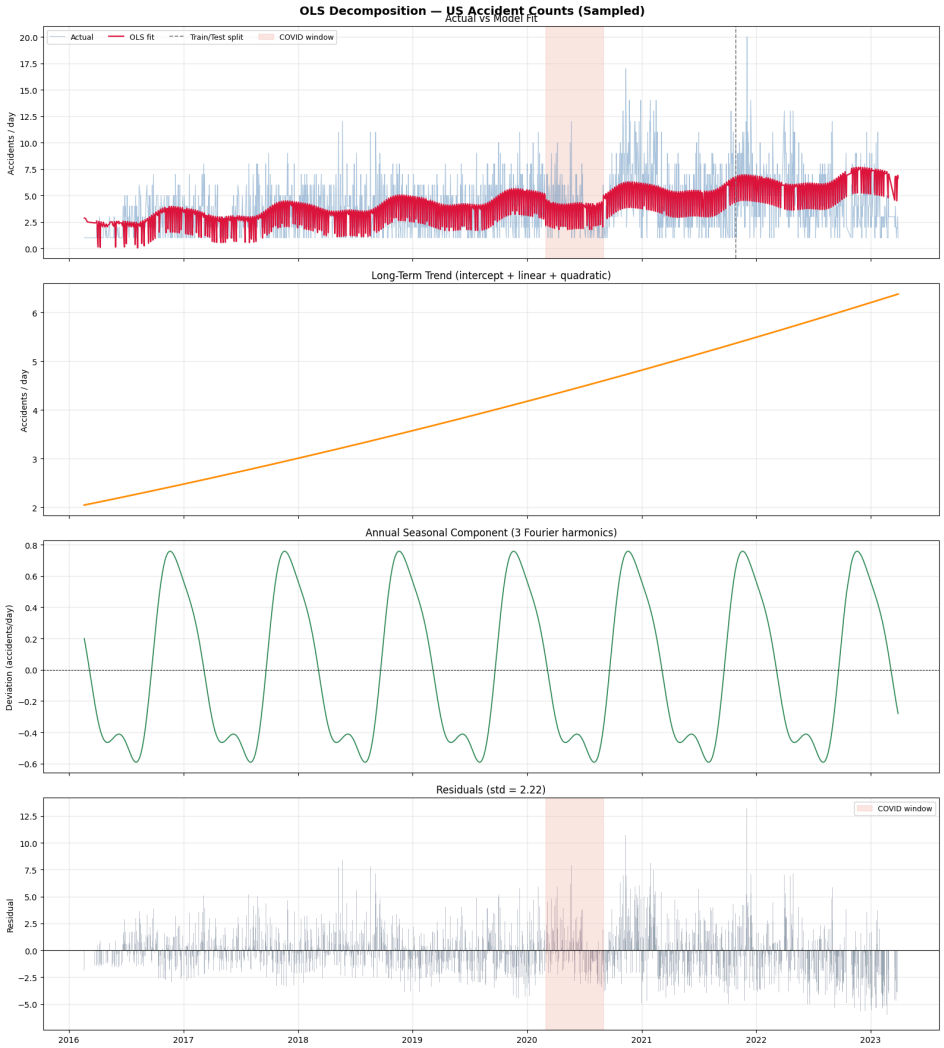

Figure 1

Figure 2 shows the learned seasonal patterns in more detail. The weekly cycle is strong and intuitive. Wednesday sees the highest accident rate at roughly 0.78 above the daily mean, while Sunday sits at the lowest point at 1.68 below the mean. This pattern reflects typical commuter traffic, with weekdays generating substantially more road activity than weekends. The annual cycle peaks around early October and troughs around mid-June, with a swing of about 1.3 accidents per day. The fall peak likely reflects the return of regular commuting patterns after summer, though the sampled dataset limits how confidently this can be interpreted.

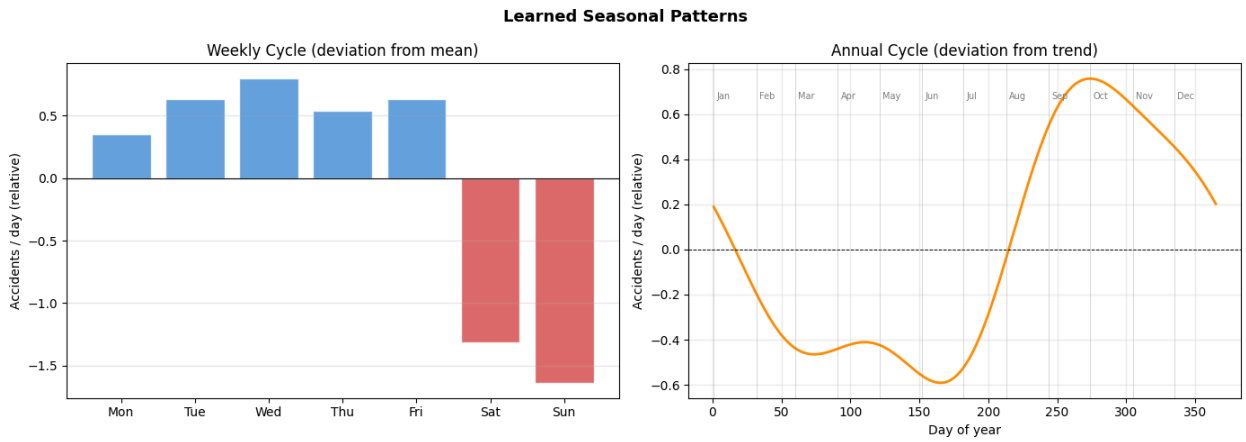

Figure 2

The AR(7) model performs better, with a test R^2 of 0.10 and a MAE of 1.99. Rather than relying on an extrapolated trend, it bases each prediction on the previous seven days of observed counts, which makes it more responsive to the current level of the series. The two most influential inputs are the previous day and the same day from the prior week, reflecting both short-term carryover and the weekly periodicity present in the data.

The SARMA(1,1)(1,1)$_7$ achieves the best test performance with an R^2 of 0.155 and a MAE of 1.94. It extends the AR approach by incorporating past forecast errors as additional inputs, which allows the model to correct for recent deviations rather than carrying them forward. The dominant predictor remains the same day from the previous week, but the negative moving average terms indicate that when recent values are unusually high, the model adjusts downward toward a more typical level. This correction is useful for count data where occasional spikes tend not to persist.

Across all three models, the consistent improvement from OLS to AR(7) to SARMA suggests there is real exploitable structure in the data. The coefficient directions across all models align with domain knowledge, and the progression in test performance reflects genuine modeling gains rather than overfitting. Applied to the full 7.7 million record dataset where daily counts would reach into the hundreds, all three models would be expected to perform substantially better as the underlying patterns would be far easier to detect.

Bayesian

The Bayesian categorical logistic regression model converged successfully across all parameters. All Rhat statistics were equal to 1, indicating that the four independent MCMC

chains produced consistent and stable estimates. The model achieved an overall classification accuracy of 48.15% on the test set. While this may not seem the best, it should be interpreted in the context of a four-class prediction problem with substantial class imbalance. A naive classifier always predicting the majority class (Severity 2) would achieve approximately 42% accuracy on the balanced test set, meaning the model offers a meaningful improvement over baseline.

The per-class performance metrics reveal important variation across severity classes. Severity 2 achieved a precision of 0.442 and recall of 0.469, with an F1 score of 0.455, indicating moderate ability to correctly identify and retrieve moderate severity accidents. Severity 3 performed best overall with a precision of 0.512, recall of 0.618, and F1 score of 0.560, suggesting the model was most capable of distinguishing serious accidents from other classes. Severity 4 showed a precision of 1.000. This means every predicted Severity 4 case was correct. But, a recall of only 0.031 and F1 of 0.061 means the model almost never predicted this class despite being perfectly accurate on the rare occasions it did. Severity 1 returned NA precision and F1 with zero recall, indicating the model never successfully predicted the least severe class. These results reflect the main challenge we faced with sparse training data for Severity 1 (81 observations) and Severity 4 (207 observations), which limited the model's ability to learn reliable decision boundaries for those classes.

Examination of the posterior coefficient distributions revealed that accident distance was the strongest and most reliable predictor of severity across all outcome classes. Distance showed credible positive effects for Severity 2, 3, and 4 relative to Severity 1. As all three credible intervals exclude zero, these effects represent the most statistically reliable findings of the model. Figure 3, which is the conditional effects plot for distance, shows this relationship clearly. At low distance values, Severity 2 dominates with approximately 40–45% predicted probability, while Severity 4 begins at around 20%. As distance increases, these probabilities invert sharply, with Severity 4 approaching near-certainty at the upper end of the observed range. This finding aligns with real-world expectations, which is that accidents involving greater road disruption are indicative of higher-speed collisions or multi-vehicle incidents, both of which would increase accident severity. This finding reinforced for us the predictive power of a Bayesian model.

Figure 4 shows Posterior coefficient distributions for all predictors across severity classes 2, 3, and 4 relative to Severity 1. Each distribution represents the posterior uncertainty around a predictor's log-odds effect. Predictors whose distributions sit clearly to the right of the dashed

zero line have a credible positive effect on that severity class. Distance is the only predictor that consistently sits to the right of zero across all three severity classes, confirming it as the dominant predictor. Temperature sits clearly to the left of zero across all classes, indicating a negative association with severity. This is an interesting finding considering higher temperatures do not technically reduce accident severity. Rather, this association likely reflects seasonal or geographical effects. Meaning that possibly warmer conditions correlate with different driving patterns, road types, or accident contexts not captured by the other predictors in the model. All remaining predictors have distributions centred on zero, indicating no reliable independent effect given the available sample size. The absence of reliable weather and visibility effects is notable given their intuitive relevance to accident severity. This most likely reflects insufficient statistical power from the small sample sizes of the minority severity classes rather than a true absence of effect in the population.

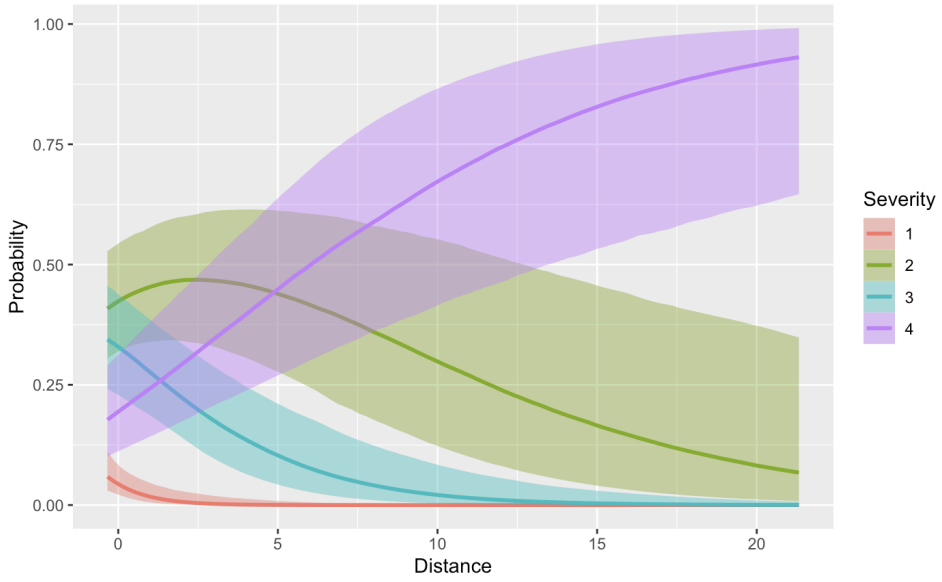

Figure 3

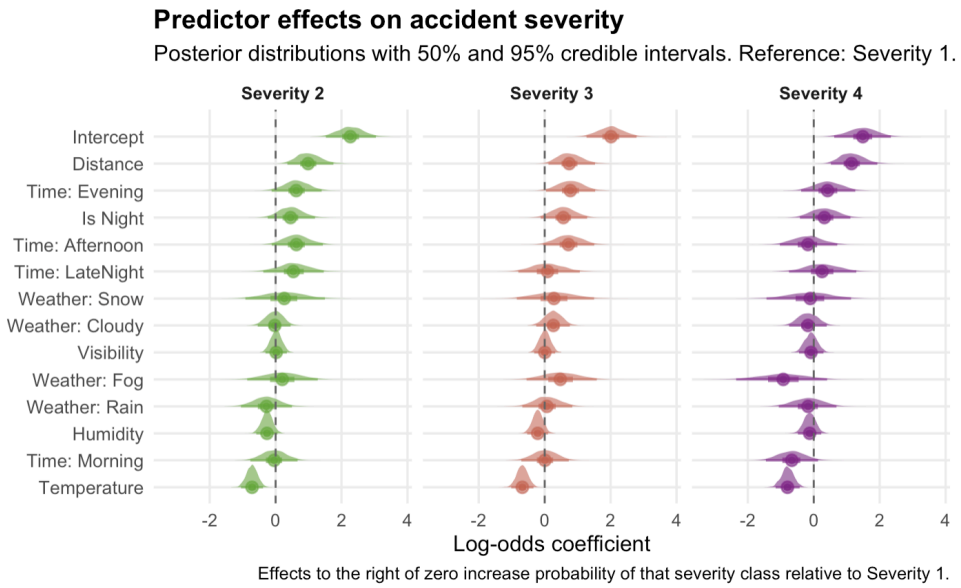

Figure 4

Conclusions and Future Work

This project applied two modeling approaches to a large-scale US traffic accident dataset to explore temporal trends in accident frequency and the factors that drive accident severity. Both components point to the same general conclusion: the structure in this data is real and recoverable, even if the sample constraints limit how cleanly it comes through.

The time series models told a consistent story. Accident frequency grew substantially over the dataset span, weekday counts reliably exceeded weekend counts, and the COVID-19 lockdown produced a measurable drop in accidents. The seasonal peak in early fall is one of the more intuitive findings, likely tied to the back to school period when commuting patterns shift and more cars return to the road at consistent times. The weekly cycle reinforces this, with Wednesday showing the highest accident rate and Sunday the lowest. The SARMA performed best out of sample, and the same-day-last-week lag being the strongest predictor in both lag-based models further supports the idea that weekly commuting rhythms are a dominant force in the data.

The Bayesian model identified accident distance as the dominant predictor of severity while finding limited evidence for the influence of environmental and temporal conditions. Performance was acceptable for the two majority classes but poor for rare severity outcomes, a limitation directly attributable to the sparsity of Severity 1 and 4 observations in the dataset.

These findings suggest that road impact distance is a meaningful signal for severity classification, while larger and more balanced datasets would be required to reliably characterise the role of weather and visibility conditions.

The biggest limitation is sample size. Working with 10,000 rows means daily counts average around four accidents per day, which introduces enough noise that even well-specified models struggle. The time series models also treated the entire country as one series, ignoring regional variation that a hierarchical model would handle better. If we continued this work, scaling to the full dataset would be the most impactful first step, followed by adding external predictors like traffic volume and exploring SARIMA to handle non-stationarity more cleanly. The weaker performance numbers reflect the sample constraints rather than a flaw in the approach, and with more data the results would be a lot cleaner.

Appendix: Implementation

https://github.com/EvelynRobert/US-Accident-Modeling-DS4420

A: Python Manual Implementation — Time Series Models

OLS Decomposition:

```
dates  = pd.to_datetime(daily_counts['Date'])
t_days = (dates - dates.iloc[0]).dt.days.values.astype(float)
t_norm = t_days / t_days.max()

n   = len(daily_counts)
dow = dates.dt.dayofweek.values

annual_feats = []
for k in range(1, 4):
    annual_feats.append(np.sin(2 * np.pi * k * t_days / 365.25))
    annual_feats.append(np.cos(2 * np.pi * k * t_days / 365.25))

weekly_feats = []
for k in range(1, 4):
    weekly_feats.append(np.sin(2 * np.pi * k * dow / 7))
```

```
    weekly_feats.append(np.cos(2 * np.pi * k * dow / 7))

covid_start = pd.Timestamp('2020-03-01')
covid_end   = pd.Timestamp('2020-08-31')
covid_dummy = ((dates >= covid_start) & (dates <=
covid_end)).astype(float).values

X = np.column_stack([
    np.ones(n),
    t_norm,
    t_norm ** 2,
    *annual_feats,
    *weekly_feats,
    covid_dummy
])
y = daily_counts['Accident_Count'].values.astype(float)

X_train = X[:split]
y_train = y[:split]

beta = np.dot(np.linalg.inv(np.dot(X_train.T, X_train)), np.dot(X_train.T,
y_train))
```

AR(7):

```
p    = 7
n_ar = len(y) - p

X_ar = np.column_stack([y[p - i - 1 : n_ar + p - i - 1] for i in range(p)])
X_ar = np.column_stack([np.ones(n_ar), X_ar])
y_ar = y[p:]

split_ar  = split - p
```

```
X_ar_train = X_ar[:split_ar]
y_ar_train = y_ar[:split_ar]

w_ar = np.dot(np.linalg.inv(np.dot(X_ar_train.T, X_ar_train)),
              np.dot(X_ar_train.T, y_ar_train))
```

SARMA(1,1)(1,1)$_7$ via Hannan-Rissanen:

```
s = 7
m = 20

X_init = np.column_stack([np.ones(len(y) - m)] +
                         [y[m - i - 1 : len(y) - i - 1] for i in range(m)])
w_init      = np.dot(np.linalg.inv(np.dot(X_init.T, X_init)), np.dot(X_init.T,
y[m:]))
eps_hat     = np.zeros(len(y))
eps_hat[m:] = y[m:] - X_init @ w_init

t0      = m + s
n_sarma = len(y) - t0

X_sarma = np.column_stack([
    np.ones(n_sarma),
    y[t0 - 1       : t0 - 1 + n_sarma],
    eps_hat[t0 - 1 : t0 - 1 + n_sarma],
    y[t0 - s       : t0 - s + n_sarma],
    eps_hat[t0 - s : t0 - s + n_sarma],
])
y_sarma = y[t0:]

split_sarma   = split - t0
X_sarma_train = X_sarma[:split_sarma]
y_sarma_train = y_sarma[:split_sarma]
```

```
w_sarma = np.dot(np.linalg.inv(np.dot(X_sarma_train.T, X_sarma_train)),
                 np.dot(X_sarma_train.T, y_sarma_train))
```

R – Bayesian Model:

```
default_prior(Severity ~ Temp + Humidity + Visibility + Distance + IsNight +
Weather_Bucket + TimeOfDay,
              data = train,
              family = categorical("logit"))

manual_prior <- c(
  brms::prior(student_t(3, 0, 5), class = "Intercept", dpar = "mu2"),
  brms::prior(student_t(3, 0, 5), class = "Intercept", dpar = "mu3"),
  brms::prior(student_t(3, 0, 5), class = "Intercept", dpar = "mu4"),
  brms::prior(normal(0, 1), class = "b", dpar = "mu2"),
  brms::prior(normal(0, 1), class = "b", dpar = "mu3"),
  brms::prior(normal(0, 1), class = "b", dpar = "mu4")
)

severity_brm <- brm(
  Severity ~ Temp + Humidity + Visibility + Distance + IsNight +
Weather_Bucket + TimeOfDay,
  family  = categorical("logit"),
  data    = train,
  chains  = 4,
  iter    = 2000,
  warmup  = 500,
  thin    = 2,
  cores   = getOption("mc.cores", 4),
  prior   = manual_prior
)
```

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